Heterogeneous Households and Aggregate Dynamics: Analytical Moment Recursion[1]

Online Appendix

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A Proofs

A.1 Proof of Proposition 1

Proposition 1 (Analytical household-level decisions)**.**
The optimal consumption, saving, and labor supply policies are:

$$c(x;X) = \frac{1-\beta}{1+\theta}(1+r(X)(1-\tau)e)a + \left(\frac{1-\beta}{1+\theta}a\xi + \kappa\right) w(X)(1-\tau)Z_L z \tag{1}$$

$$a'(x;X) = \beta a\left((1+r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right) + T(X) \tag{2}$$

$$n(x;X) = \max\left\{\kappa(1+\theta) + \xi a - \frac{\theta\, c(x;X)}{w(X)(1-\tau)Z_L z},\ 0\right\} \tag{3}$$

Proof.
I assume the constraint $a' \geq 0$ does not bind and later verify that the solution satisfies this. From the first-order conditions,

$$[c]: \quad \frac{1}{c - \kappa w(X)(1-\tau)Z_L z} = \lambda \tag{4}$$

$$[a']: \quad \beta\,\mathbb{E}\left[v_a(x';X')\right] = \lambda \tag{5}$$

$$[n]: \quad \frac{\theta}{\kappa + \xi a - n} = \lambda w(X)(1-\tau)Z_L z. \tag{6}$$

From the envelope condition,

$$[a]: \quad v_a(x;X) = \frac{\theta\xi}{\kappa + \xi a - n} + \lambda(1 + r(X)(1-\tau)e). \tag{7}$$

From Equations (4) and (6),

$$\frac{\theta}{\kappa + \xi a - n} = \frac{w(X)(1-\tau)Z_L z}{c - \kappa w(X)(1-\tau)Z_L z}. \tag{8}$$

Rearranging,

$$n(x;X) = \kappa(1+\theta) + \xi a - \theta\frac{c(x;X)}{w(X)(1-\tau)Z_L z}. \tag{9}$$

The budget constraint can be written as:

$$(1+\theta)(c - \kappa w(X)(1-\tau)Z_L z) + a' = ((1 + r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi)a + T(X). \tag{10}$$

The transfer $T(X)$ enters savings as an additive intercept. For the multiplicative component, conjecture $v(a, z, e; X) = A(z, e, X) \log a + B(z, e, X)$. Under log utility, the Euler equation combined with the conjectured form yields saving share β (the current discount factor from X)—the standard result of Brock and Mirman (1972). Verification: substituting back confirms $A(z, e, X) = (1 + \theta)/(1 - \beta)$ (independent of beliefs about future aggregates, the basis for Theorem 1; see Appendix A.8 for the general ζ recursion when β varies) and B satisfies a well-defined recursion. Since $a' = \beta a(\cdot) + T \geq T > 0$, the borrowing constraint is always slack. ■

A.2 Proof of Proposition 2

Proof sketch. The saving rate $s(x; X) = (a'(x; X) - a)/y(x; X)$ inherits the properties of the saving policy a'. Property (ii) follows immediately from $s(0; X) = T/(\kappa\theta w(X)(1 - \tau)Z_L z + T) > 0$. Property (i) follows from taking the limit $a \to \infty$, under which the transfer T becomes negligible. Property (iii)—monotonicity and concavity—follows from direct differentiation: $\partial s/\partial a > 0$ because the multiplicative component of a' grows faster than income, and $\partial^2 s/\partial a^2 < 0$ because the subsistence floor absorbs a diminishing share of resources. ■

A.3 Proof of Proposition 3

Proof. The saving policy $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L\xi z) + T$ is linear in a with coefficients that depend on (z, e) and current prices. Aggregating across the distribution:

For aggregate capital: $K' = \int a'(x; X)\, d\Phi = \beta[(1+\tilde{r}+\tilde{w}Z_L\xi)K+\tilde{r}\,\mathrm{cov}(e, a)+\tilde{w}Z_L\xi\,\mathrm{cov}(z, a)]+T$, using $\mathbb{E}[e] = \mathbb{E}[z] = 1$.

For $\mathrm{cov}(z', a')$: the idiosyncratic processes follow stationary Markov chains (discretized via Tauchen's method from AR(1) processes in levels: $z' - 1 = \rho_z(z - 1) + \varepsilon'_z$, and analogously for e). Since a' is linear in a and ε'_z is independent of current states, the covariance expands as $\mathbb{E}[z'a'] - \mathbb{E}[z']\mathbb{E}[a']$. Using the conditional linearity $\mathbb{E}[z'|z] = (1 - \rho_z) + \rho_z z$ and the fact that, under the stationary distribution, $\mathbb{E}[z^2 a] \approx \sigma_z^2 K + (1 + \sigma_z^2/\mathrm{var}(z))\,\mathrm{cov}(z, a) \cdot K/K$ (by the linear projection of a on z), the covariance recursion becomes $\mathrm{cov}(z', a') = \rho_z\phi_z\beta[\tilde{w}Z_L\xi\sigma_z^2 K + (1 + \tilde{r} + \tilde{w}Z_L\xi)\mathrm{cov}(z, a)]$, where $\phi_z = \sigma_z^2/(\sigma_z^2 + \sigma_{\varepsilon,z}^2)$ is the signal-to-noise ratio governing the speed of distributional reshuffling.

An analogous computation yields the recursion for $\mathrm{cov}(e', a')$, and the independence of z and e produces the zero off-diagonal entries. The system closes because prices (r, w) depend on Φ only through $(K, \mathrm{cov}(z, a), \mathrm{cov}(e, a))$ via the labor supply equation, which itself depends only on these three moments. Uniqueness follows from the contraction property of

the moment map when $\rho(\beta\widetilde{\mathcal{G}}) < 1$. ■

A.4 Proof of Corollary 2

Proof. The system $\mathcal{M}' = \beta\widetilde{\mathcal{G}}\mathcal{M} + [T, 0, 0]'$ is a non-homogeneous linear recursion. When the spectral radius $\rho(\beta\widetilde{\mathcal{G}}) < 1$, the matrix $I - \beta\widetilde{\mathcal{G}}$ is invertible and the unique fixed point $\mathcal{M}^* = (I - \beta\widetilde{\mathcal{G}})^{-1}[T, 0, 0]'$ is globally attracting. This follows from the standard theory of linear difference equations: the general solution is $\mathcal{M}_t = (\beta\widetilde{\mathcal{G}})^t(\mathcal{M}_0 - \mathcal{M}^*) + \mathcal{M}^*$, which converges to $\mathcal{M}^*$ as $t \to \infty$ whenever all eigenvalues of $\beta\widetilde{\mathcal{G}}$ lie inside the unit circle. The additive component $T > 0$ is essential: it places the system in the class of Kesten processes (Kesten, 1973), which require both multiplicative and additive components to generate a well-defined stationary distribution with Pareto tails. Without T, the system is purely multiplicative and does not admit a non-degenerate stationary solution. ■

A.5 Proof of Corollaries 3 and 4

Proof of Corollary 3. Set $e = 1$ throughout. Then $\mathrm{cov}(e, a) = 0$ for all t, the third row and column of $\widetilde{\mathcal{G}}$ become irrelevant, and the system reduces to the 2×2 recursion stated. ■

Proof of Corollary 4. Set $e = 1$ and $z = 1$. Then both covariances vanish identically, and the system reduces to a single equation $K' = \beta(1 + \tilde{r} + \tilde{w}Z_L\xi)K + T$. The aggregate consumption function follows from $C = Y + (1 - \delta)K - K'$. ■

A.6 Proof of Theorem 1

Proof. The saving policy $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}Z_L\xi z) + T$ depends only on current after-tax prices $(\tilde{r}, \tilde{w})$, current Z_L, the current transfer T, and the household's current state (a, z, e). None of these objects involves expectations of future aggregate states. The consumption policy $c(x; X)$ and labor supply $n(x; X)$ inherit this property. Since all individual policies are independent of beliefs about future aggregates, the aggregate law of motion $\mathcal{M}' = \beta\widetilde{\mathcal{G}}\mathcal{M} + [T, 0, 0]'$ is also independent of beliefs: it depends only on current prices (which are functions of the current state $\mathcal{M}$) and the current transfer.

The mechanism is the certainty-equivalence property of log preferences: income and substitution effects of future return changes exactly cancel, so saving depends only on current resources and prices. This aggregates because linear policies preserve the moment structure. ■

A.7 Proof of Proposition 4

Proof. The BRCE requires that household policies be optimal given *arbitrary* beliefs about aggregate dynamics. Theorem 1 establishes that the saving policy is belief-invariant. Since the aggregate law of motion depends only on the collection of saving policies (through the moment aggregation), and since these policies are identical regardless of beliefs, the BRCE law of motion coincides with the MRCE law of motion. Uniqueness of the MRCE then implies uniqueness of the BRCE, and the allocations are identical. ■

A.8 Saving share under time-varying β

Derivation. Since β is an element of the aggregate state X and may vary over time, the Bellman equation for the multiplicative component of wealth (ignoring the additive transfer T) is:

$$v_t(a, z, e; X) = \max_{a'} (1+\theta) \log \tilde{c} + \beta_t \, \mathbb{E}_t \left[v_{t+1}(a', z', e'; X') \right], \tag{11}$$

where $\tilde{c} \equiv c - \kappa w(1-\tau) Z_L z$ is consumption net of the subsistence floor. Conjecture $v_t(a, z, e; X) = \zeta_t \log a + B_t(z, e, X)$, where ζ_t may depend on aggregate conditions but not on individual wealth.

From the envelope condition, $v_{a,t} = \zeta_t / a$. Combined with the first-order conditions (as in Appendix A.1), the envelope condition yields $\zeta_t / a = \lambda_t \Pi_t$, where $\Pi_t = (1 + \tilde{r}_t e) + \tilde{w}_t Z_L \xi z$ is the gross multiplicative return on wealth. Since $\lambda_t = 1/\tilde{c}_t$, this gives $\tilde{c}_t = a \Pi_t / \zeta_t$, and the budget constraint implies:

$$a' = \Pi_t \, a \left(1 - \frac{1+\theta}{\zeta_t} \right). \tag{12}$$

The first-order condition for a' requires $\beta_t \, \mathbb{E}_t[\zeta_{t+1}/a'] = \zeta_t / (a \Pi_t)$. Substituting the expression for a' and simplifying:

$$\zeta_t = (1+\theta) + \beta_t \, \mathbb{E}_t[\zeta_{t+1}]. \tag{13}$$

The saving share out of the multiplicative component is therefore:

$$s_t = 1 - \frac{1+\theta}{\zeta_t} = \frac{\beta_t \, \mathbb{E}_t[\zeta_{t+1}]}{\zeta_t}. \tag{14}$$

Constant β. When β is constant, $\zeta = (1+\theta)/(1-\beta)$ and $s = \beta$, recovering the steady-state result.

Belief invariance. Under the BRCE, households treat the current β as permanent. The recursion (13) then gives $\zeta = (1+\theta)/(1-\beta)$, so $s = \beta$. This is a static function of the current discount factor—no expectations are required. The saving policy is $a'(x; X) = \beta\, a\Pi + T$, and the moment recursion is $\mathcal{M}' = \beta\, \widetilde{\mathcal{G}}\, \mathcal{M} + [T, 0, 0]'$.

Time-varying τ_t. The tax rate enters only through after-tax prices in the income base Π_t, and through the transfer T_t (which is an MIT shock). Since τ_t does not appear in the ζ recursion (13), time-varying flat tax rates preserve the moment-recursive structure without any additional state variables.

Separate capital and labor tax rates. The same logic extends to separate time-varying tax rates $\tau_{K,t}$ and $\tau_{L,t}$. After-tax factor prices become $\tilde{r}_t = r_t(1 - \tau_{K,t})$ and $\tilde{w}_t = w_t(1 - \tau_{L,t})$, and the transfer is $T_t = \eta\big[\tau_{K,t}(r_t + \delta)K_t + \tau_{L,t}\, w_t L_t\big]$. Because both rates enter multiplicatively on factor incomes, the household's next-period wealth $a' = s_t \cdot a\Pi_t + T_t$ remains *linear* in individual wealth a—the key property that preserves the moment-recursive structure. In particular, the saving share $s_t = \beta_t$ is still independent of $(\tau_{K,t}, \tau_{L,t})$: tax rates shift *what* is saved (through Π_t and T_t) but not the *share* of wealth saved. Progressive taxes or tax brackets, by contrast, would make the effective marginal rate depend on household-level income $a\Pi_t$, breaking linearity in a and destroying the MRCE's closed-form aggregation. ■

B Extended Theoretical Discussion

B.1 Transition matrix interpretation

The MRCE state vector has three components: aggregate capital K (the mean of the wealth distribution), $\mathrm{cov}(z, a)$ (the alignment between labor productivity and wealth), and $\mathrm{cov}(e, a)$ (the alignment between return heterogeneity and wealth). The first is the single state that Krusell and Smith (1998) found nearly sufficient numerically.

The law of motion $\mathcal{M}' = \beta\, \widetilde{\mathcal{G}}\, \mathcal{M} + [T, 0, 0]'$ is a non-homogeneous linear recursion. The diagonal entries show that each moment feeds into its own future value. The off-diagonal entries capture cross-moment interactions: the $(1, 3)$ entry $\tilde{r}$ shows that return-wealth sorting contributes to future capital through extra saving by high-return wealthy households, while the $(3, 1)$ entry $\phi_e \rho_e \tilde{r} \sigma_e^2$ shows that higher capital strengthens return-wealth sorting through higher returns.

The zero $(2, 3)$ and $(3, 2)$ entries reflect the independence of z and e: the two covariances do not directly interact. Each interacts with aggregate capital, but the productivity-wealth and return-wealth sorting channels operate separately, connected only through their common

dependence on K.

B.2 Signal-to-noise ratios

The signal-to-noise ratios ϕ_z and ϕ_e control the rate at which idiosyncratic shocks reshuffle households across the distribution. High ϕ (close to one) means persistent shocks and slow reshuffling: sorting between productivity and wealth strengthens over time. Low ϕ (close to zero) means rapid reshuffling: the covariance between productivity and wealth dissipates quickly. In $\widetilde{\mathcal{G}}$, ϕ scales the off-diagonal entries, governing the speed of distributional evolution. Economies with persistent idiosyncratic shocks exhibit slow-moving distributional states that interact strongly with aggregate fluctuations; economies with transitory shocks have fast-adjusting distributions that play a smaller aggregate role.

B.3 Special cases: extended discussion

Approximate aggregation. When there is no return heterogeneity ($e = 1$), the system reduces to two states: K and $\mathrm{cov}(z, a)$. This is the analytical counterpart of the Krusell–Smith economy. Krusell and Smith (1998) found numerically that the mean of the wealth distribution is "nearly sufficient" to forecast prices. The present result shows that the mean is *exactly* sufficient if and only if $\mathrm{cov}(z, a) = 0$. In practice, this covariance is small relative to K in many calibrations, explaining why approximate aggregation works well numerically. But the error grows with the persistence and dispersion of idiosyncratic productivity—precisely the conditions under which the Krusell–Smith approximation breaks down.

Representative-agent nesting. When all heterogeneity is eliminated, both covariances vanish and the model reduces to a stochastic growth model with log preferences and endogenous labor. The aggregate consumption function $C = \frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \kappa\tilde{w}Z_L$ has a permanent-income term proportional to wealth and a subsistence term proportional to the wage. The heterogeneous-agent model adds distributional corrections that, while small in normal times, become significant during episodes of high inequality.

B.4 State-dependent inequality dynamics

The transition matrix $\widetilde{\mathcal{G}}$ makes inequality feedback precise. The off-diagonal entries $\phi_e\rho_e\tilde{r}\sigma_e^2$ and $\phi_z\rho_z\tilde{w}Z_L\xi\sigma_z^2$ show that higher capital strengthens both return-wealth and productivity-wealth sorting, generating endogenous amplification: a positive shock raises capital, which raises returns, which strengthens sorting, which raises future capital.

This "wealth spiral" is consistent with the secular rise in top wealth shares documented by Piketty (2014) and Saez and Zucman (2016). The rate of concentration is governed by the dominant eigenvalue of $\beta\widetilde{\mathcal{G}}$; the steady-state level is pinned down by the fixed point $\mathcal{M}^*$. Counteracting forces limit concentration: consumption ($\beta < 1$ ensures a fraction $1-\beta$ of wealth is consumed each period) and the transfer T, which provides an additive floor benefiting low-wealth households disproportionately.

C Asset Pricing Details

The model delivers a novel asset pricing result: individual return heterogeneity, traditionally viewed as diversifiable, carries a positive risk premium because it has aggregate consequences (Constantinides and Duffie, 1996; Storesletten et al., 2004).

In the standard complete-markets framework, idiosyncratic risk washes out by the law of large numbers. Here it does not, because return heterogeneity creates systematic wealth sorting: households with persistently high e accumulate more wealth, work less through the wealth effect, and thereby depress aggregate labor supply and returns for *everyone*. Because e affects aggregate outcomes through $\text{cov}(e, a)$, it is not fully diversifiable.

C.1 Static and dynamic decomposition

The asset pricing implications decompose into a static and a dynamic component.

Static effect. Holding the aggregate state fixed, higher $\text{cov}(e, a)$ reduces the current return:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L}\frac{\partial L}{\partial \text{cov}(e, a)} < 0 \tag{15}$$

Higher sorting strengthens the wealth effect on labor supply, reducing employment and the marginal product of capital.

Dynamic effect. One period ahead, the effect is reinforced:

$$\frac{\partial r(\mathcal{M}', A')}{\partial \text{cov}(e, a)} = \frac{\partial r'}{\partial K'}\frac{\partial K'}{\partial \text{cov}(e, a)} + \frac{\partial r'}{\partial \text{cov}(e', a')}\frac{\partial \text{cov}(e', a')}{\partial \text{cov}(e, a)} < 0 \tag{16}$$

Both terms are negative: higher sorting today raises capital accumulation (depressing future returns through diminishing marginal products) and propagates into tomorrow's sorting (continuing to depress returns through labor supply).

C.2 Wealth mobility

The model provides an analytical half-life of wealth:

$$T(\mathcal{M}, A) = \frac{\log(2)}{\log(\beta(1 + r(\mathcal{M}, A)(1 - \tau)))} \tag{17}$$

Faster mean reversion of $\text{cov}(e, a)$ implies quicker distributional adjustment, but slow rank mobility means individual wealth positions change gradually. The combination generates persistent aggregate effects of idiosyncratic return shocks. These results suggest that models with heterogeneous returns (Fagereng et al., 2020; Bach et al., 2020) have implications for asset prices and aggregate dynamics, not just the wealth distribution.

C.3 Risk premium sensitivity

To quantify the pricing channel, we compare steady-state economies at different σ_e. At baseline ($\sigma_e = 0.01$), the interest rate effect is -0.001 basis points—negligible. The premium grows quadratically:

$$\underbrace{-0.03 \text{ bp}}_{\sigma_e=0.05} \quad \underbrace{-0.10 \text{ bp}}_{\sigma_e=0.10} \quad \underbrace{-0.70 \text{ bp}}_{\sigma_e=0.50} \quad \underbrace{-1.01 \text{ bp}}_{\sigma_e=0.83}$$

The threshold for a 1 basis point premium requires $\sigma_e \approx 0.83$. At empirically realistic σ_e, the main consequence of return heterogeneity operates through the *amplification channel* (state-dependent impulse responses), not the *pricing channel*.

D Epstein-Zin Extension

Under Epstein-Zin preferences with unit elasticity of intertemporal substitution (EIS = 1), the flow composite is:

$$\tilde{c} \equiv c - \kappa Z_L z w(X)(1 - \tau), \qquad m \equiv (\kappa + \xi a) - n, \qquad X^{\text{comp}} \equiv \tilde{c}\, m^{\theta}.$$

The intratemporal optimality condition implies $m = (\theta/(w(X)(1 - \tau)Z_L z))\tilde{c}$, so:

$$X^{\text{comp}} = \tilde{c}^{1+\theta} \left(\frac{\theta}{w(X)(1 - \tau)Z_L z} \right)^{\theta}.$$

The recursive value function satisfies:

$$\log V_t = (1-\beta)\log X_t + \beta \log\left(\mathbb{E}_t\left[V_{t+1}^{1-\gamma}\right]\right)^{\frac{1}{1-\gamma}}, \qquad \beta \in (0,1), \quad \gamma > 0.$$

The critical result is that, with EIS = 1, the optimal saving share $s_a(s)$ satisfies the same type of fixed-point condition as in the log case:

$$s_a(s) = \left[1 + \left(\frac{\beta(1+\theta)^p\,\bar{B}(s)}{C(s)}\right)^{\frac{1}{p-1}}\right]^{-1},$$

where $p \equiv (1+\theta)(1-\sigma)$, $C(s) \equiv (\theta/(w(X)(1-\tau)Z_L z))^{\theta(1-\sigma)}$, and $\bar{B}(s) \equiv \mathbb{E}[B(s') \mid s]$ is a conditional expectation computed from the fixed-point system:

$$B(s) = \Pi(s)^p\left[C(s)\left(\frac{1-s_a(s)}{1+\theta}\right)^p + \beta\, s_a(s)^p\,\bar{B}(s)\right],$$

with $\Pi(s) \equiv (1 + r(X)(1-\tau))e + w(X)(1-\tau)Z_L z\xi$.

The MRCE and BRCE continue to hold under Epstein-Zin preferences with unit EIS. The saving share remains independent of wealth, and the law of motion retains its closed-form matrix structure. The parameter γ controls risk aversion over consumption *lotteries* (cross-sectional), while β and θ control intertemporal and intratemporal tradeoffs (time-series). This separation allows the model to match both the equity premium (through γ) and the saving rate (through β). With $\gamma \gg 1$, the wealth-sorting mechanism generates a substantially larger risk premium than under log utility.

E Additional Quantitative Results

E.1 Implied TFP comparison

Figure E.1 compares implied TFP from the MRCE (accounting for distributional covariances) with TFP from a representative-agent version (all covariances set to zero). The two series track closely at business cycle frequencies, but the HA model implies a slightly different level, reflecting distributional sorting's contribution to labor supply. The Fernald utilization-adjusted TFP series serves as an external benchmark.

E.2 Structural interpretation of $Z_{L,t}$

The aggregate labor productivity $Z_{L,t}$, introduced as a model primitive in Section 2, is a structural object—not a reduced-form residual. Three properties distinguish it from reduced-form labor wedges.

Model primitive with distributional consequences. $Z_{L,t}$ scales household effective earnings $\tilde{w}_t Z_{L,t} z$ without entering the production function. It appears in three places in the moment recursion:

1. The effective wage in household labor supply: $\tilde{w}_t Z_{L,t}$,
2. The wealth-leisure complementarity: $\tilde{w}_t Z_{L,t} \tilde{\xi}_t$, and
3. The transition matrix $\widetilde{\mathcal{G}}_t$, which governs the evolution of distributional covariances.

Through channel (3), a decline in $Z_{L,t}$ reduces the rate at which productivity-wealth sorting accumulates, altering the dynamics of wealth inequality. This distinguishes $Z_{L,t}$ from a Chari et al. (2007) labor wedge, which operates only through the intratemporal margin and has no distributional content.

Asymmetric entry. $Z_{L,t}$ enters household decisions but *not* the production function $Y_t = A_t K_t^{\alpha} L_t^{1-\alpha}$, which uses raw hours. This asymmetry means TFP (A_t) is *directly identified* from the production function—no fixed-point computation is required. $Z_{L,t}$ then absorbs the discrepancy between model-implied and observed hours conditional on A_t, β_t, and the distributional state. The asymmetry is testable: a shift in $Z_{L,t}$ changes hours and investment composition without altering technology.

External validation. Section 4.2 reports strong correlations between $Z_{L,t}$ and labor market indicators not used in the inversion: the employment-population ratio (level $\rho = 0.70$, HP-cycle $\rho = 0.62$), the labor force participation rate (0.58, 0.42), and average weekly manufacturing hours (0.63, 0.33). Only GDP, the capital stock, and per-capita hours enter the inversion. The employment-population ratio captures the extensive margin of labor utilization—exactly the margin $Z_{L,t}$ reflects.

Equally informative is the *negative* HP-cycle correlation ($\rho = -0.70$) with the BLS labor composition index (MPU4900092, 1987–2019), which measures workforce quality through demographics. In recessions, low-productivity workers are disproportionately separated, raising average quality—the "cleansing" effect of Caballero and Hammour (1994). Meanwhile, $Z_{L,t}$ declines through underutilization, matching deterioration, and foregone on-the-job learning.

The sign reversal confirms that $Z_{L,t}$ captures a *utilization* margin distinct from demographic composition.

Comparison with the CKM labor wedge. In Chari et al. (2007), the labor wedge $1 - \tau_L$ measures the gap between the intratemporal optimality condition and data but does not identify its source. In contrast, $Z_{L,t}$ has a specific structural role: it is the mean of the idiosyncratic productivity distribution, and its variation maps to composition, matching, and utilization channels measurable independently. The external validation above confirms this structural content is empirically relevant, not merely a relabeling of the CKM residual.

E.3 Aggregate dynamics

This appendix provides the full details of the aggregate dynamics results summarized in Section 4.3.

Investment decomposition. Prediction **P1** (approximate aggregation) can be tested directly. Investment decomposes as:

$$I^{\text{rep}} = \beta(1 + \tilde{r} + \tilde{w} Z_L \xi)K + T - (1 - \delta_t)K \qquad \text{(representative-agent channel)} \tag{18}$$

$$I^{\text{ret}} = \beta \tilde{r} \operatorname{cov}(e, a) \qquad \text{(return-sorting channel)} \tag{19}$$

$$I^{\text{prod}} = \beta \tilde{w} Z_L \xi \operatorname{cov}(z, a) \qquad \text{(productivity-sorting channel)} \tag{20}$$

Figure E.2 plots the three channels. The representative-agent channel accounts for over 99.9 percent of mean investment and over 100 percent of its covariance with total investment, confirming analytically what Krusell and Smith (1998) found numerically. The distributional channels each contribute less than 0.01 percent of mean investment, yet they generate the state-dependent amplification that a representative-agent model cannot produce.

Note that model-implied investment $I_t^{\text{model}} = K_{t+1} - (1 - \delta_t)K_t$ need not coincide with measured gross investment from the national accounts. The model abstracts from inventory investment, government investment, and residential construction included in NIPA aggregates.

State-dependent dynamics. To test prediction **P2** (state-dependent amplification), I compute generalized impulse response functions (GIRFs) for a 2 percent negative TFP shock launched from two initial states: the 1952 distribution (low inequality, $\operatorname{cov}(e, a) = 0.001$) and the 2019 distribution (high inequality, $\operatorname{cov}(e, a) = 0.006$). Following Koop et al. (1996), each GIRF averages over 1,000 stochastic TFP paths with common innovations. The impact

effect on output is 32 percent larger from 2019 (-2.64% vs. -2.01%), and the cumulative 10-year loss is 64 percent larger (-30.7% vs. -18.7%).

The mechanism operates through three reinforcing channels. First, the *exposure channel*: the return-sorting term $\beta\tilde{r}\,\mathrm{cov}(e,a)$ in the capital recursion means that aggregate investment depends on the covariance between returns and wealth. When $\mathrm{cov}(e,a)$ is large, a larger share of saving originates from households most sensitive to the aggregate return $\tilde{r}$, amplifying the investment contraction.

Second, the *labor withdrawal channel*: the wealth-effect term $-\frac{\theta(1-\beta)}{(1+\theta)\tilde{w}Z_L}\tilde{r}\,\mathrm{cov}(e,a)$ shows that return-wealth sorting reduces aggregate labor supply. When $\mathrm{cov}(e,a)$ is large, wealthy households withdraw more labor through wealth-leisure complementarity ($\kappa+\xi a$), depressing output beyond the direct capital channel.

Third, the *persistence channel*: the off-diagonal entry $\widetilde{\mathcal{G}}_{31} = \phi_e\rho_e\tilde{r}\sigma_e^2$ feeds capital back into future return-wealth sorting. Because the dominant eigenvalue of $\beta\widetilde{\mathcal{G}}$ exceeds unity along the historical path, the distributional state does not mean-revert: sorting that amplified the initial shock *persists*, keeping the economy on a lower trajectory. This is why the gap in cumulative losses (64 percent) far exceeds the gap in impact effects (32 percent). Figure E.3 displays the GIRFs.

Return risk premium. The interest rate effect of return-wealth sorting is -0.001 basis points at $\sigma_e = 0.01$—negligible. The premium grows quadratically with σ_e but requires $\sigma_e \approx 0.83$ to reach 1 basis point. This identifies the amplification channel, not the pricing channel, as the economically significant consequence of return heterogeneity at realistic calibrations. The full sensitivity analysis appears in Appendix C.3.

Counterfactual analysis. With three structural shocks, I compare the baseline (all historical shocks) with three counterfactual economies—fixing β_t, fixing $Z_{L,t}$, or fixing both—to isolate each shock's marginal contribution. Figure E.4 compares log output paths. Three findings stand out. First, fixing $Z_{L,t}$ at its sample mean ($\bar{Z}_L = 1.08$) yields a covariance share of $+121$ percent—$Z_{L,t}$ is the dominant driver of cyclical output variation. Second, fixing β_t at its sample mean ($\bar{\beta} = 0.954$) yields -60 percent—patience partially offsets output cycles. Third, fixing both (TFP only) yields a std ratio of 0.45 against the baseline, indicating TFP alone understates cyclical amplitude.

TFP is the dominant driver of wages and the long-run interest rate, $Z_{L,t}$ is the primary driver of hours fluctuations, and β_t partially offsets TFP-driven investment cycles through countercyclical patience.

E.4 Counterfactual multi-variable panels

Figure E.5 extends the counterfactual comparison to output, labor supply, the interest rate, and the wage.

E.5 Transition dynamics

Figure E.6 plots the dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$ over time. The dominant eigenvalue has mean 1.017 and exceeds unity in 99 percent of sample years, underscoring the extreme persistence of distributional dynamics. The near-unit-root behavior generates endogenous persistence in aggregate dynamics that converts transitory shocks into persistent output movements.

E.6 Distributional state dynamics

Figure E.7 plots the two covariances and their capital-normalized counterparts. The productivity-wealth covariance $\mathrm{cov}_t(z, a)$ increases from 0.003 in 1950 to 0.024 in 2019 (a factor of 7.2), while $\mathrm{cov}_t(e, a)$ grows from 0.0005 to 0.003. Both are procyclical. The normalized ratio $\mathrm{cov}(e, a)/K$ remains nearly constant, reflecting the near-unit-root dynamics of e.

E.7 Sorting decomposition

Figure E.8 reports the analytical decomposition of distributional sorting into innovation and persistence channels. Persistence dominates: on average, 84 percent of $\mathrm{cov}(z, a)$ growth comes from the persistence channel. The right panel confirms that the new sorting rate $\widetilde{\mathcal{G}}_{21}$ is strongly positively correlated with TFP, providing a structural foundation for the TFP-inequality nexus.

E.8 Historical distribution of TFP gains

Table E.1 reports decade averages of TFP gain shares by wealth decile. Figure E.9 visualizes the evolving distribution. The top decile captures approximately 70 percent of TFP-driven wealth gains throughout the sample.

E.9 Consumption dynamics

The MRCE delivers closed-form aggregate consumption:

$$C_t = \frac{1-\beta_t}{1+\theta}\left[(1+\tilde{r}_t+\tilde{w}_t Z_{L,t}\xi_t)K_t + \tilde{r}_t\,\mathrm{cov}_t(e,a) + \tilde{w}_t Z_{L,t}\xi_t\,\mathrm{cov}_t(z,a)\right] + \kappa\tilde{w}_t Z_{L,t}.$$

Constant marginal propensity to consume. The household MPC out of a one-time transfer is $(1-\beta_t)/(1+\theta)$, independent of wealth. At baseline ($\beta = 0.96$, $\theta = 2$), the annual MPC is 1.3 percent, reaching 6.6 percent after five years and 26.3 percent after twenty. The constant MPC follows directly from linear policy functions, contrasting with models where borrowing-constrained households exhibit MPCs of 0.3–0.5 (Kaplan et al., 2018).

Procyclical MPC. The time-varying MPC $(1-\beta_t)/(1+\theta)$ is procyclical: when β_t is low (less patience), the MPC is higher. Under the three-shock specification, mean $\beta_t = 0.954$ (std 0.010) yields an average MPC of 1.5 percent.

E.10 Historical episode analysis

The three-shock inversion provides a structural narrative for each recession. At annual frequency, three mild recessions (1960–61, 1969–70, 2001) produce negligible output declines and are omitted. The full classification appears in Section 4.6 and Table 6. Figure E.10 shows shock paths for two contrasting episodes.

E.11 Cross-sectional distributional dynamics

We simulate 500,000 agents along the historical path (1950–2019) using the three-shock inversion results. Each agent i carries state (a_{it}, z_{it}, e_{it}), where z and e follow AR(1) processes with $\sigma_z = 0.12$, $\sigma_e = 0.01$ at baseline. Countercyclical dispersion amplifies these: $\sigma_{z,t} = 1.75 \times \sigma_z$ when $Z_{L,t}$ declines (Storesletten et al., 2004), and $\sigma_{e,t} = 4.1 \times \sigma_e$ when $\Delta Y_t < 0$ (Bloom et al., 2018). Wealth evolves as $a_{it+1} = \beta_t R_{it} a_{it} + T_t$ where $R_{it} = 1 + \tilde{r}_t e_{it} + \tilde{w}_t Z_{L,t} \tilde{\xi}_t z_{it}$, and consumption is $c_{it} = \frac{1-\beta_t}{1+\theta} R_{it} a_{it} + \kappa \tilde{w}_t Z_{L,t} z_{it}$.

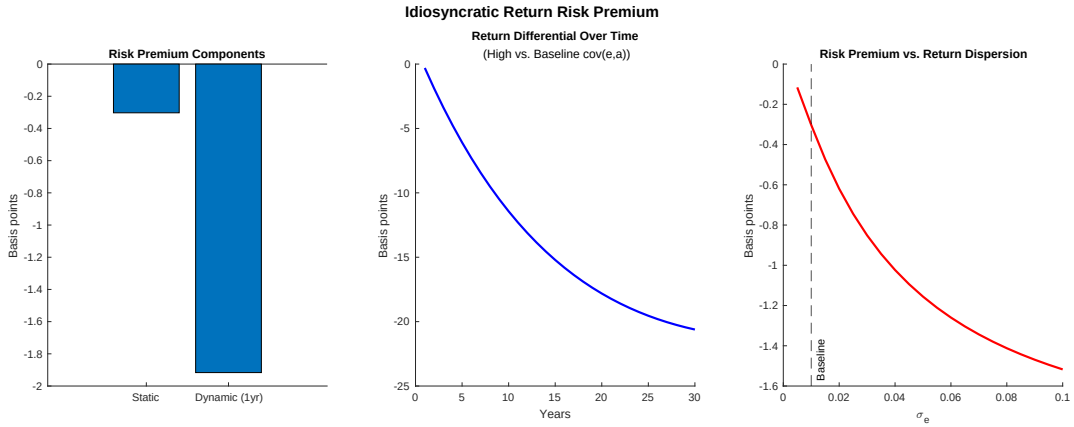
The simulation begins after a 2,000-period burn-in at steady-state prices with GE feedback (recomputing r, w from $K_t = \mathbb{E}[a_{it}]$ each period) and wealth truncation at $\bar{a} = 200$. This allows the Kesten process to develop its Pareto tail; post-burn-in wealth Gini is 0.827. During the historical simulation, the truncation boundary scales with capital ($\bar{a}_t = (\bar{a}/K_{\text{ss}}) \cdot K_t$) to maintain a constant max-to-mean ratio.

Distributional moments by recession appear in Table 5. Three patterns emerge: (i) consumption inequality rises modestly in recessions (+1.4%) while income CV falls (−3.7%); (ii) the Great Recession produces the highest consumption dispersion (CV 1.103, +42% above normal), reflecting falling β_t interacting with high wealth concentration; (iii) consumption kurtosis ranges from 33.7 (1957–58) to 90.3 (2007–09).

E.12 Robustness

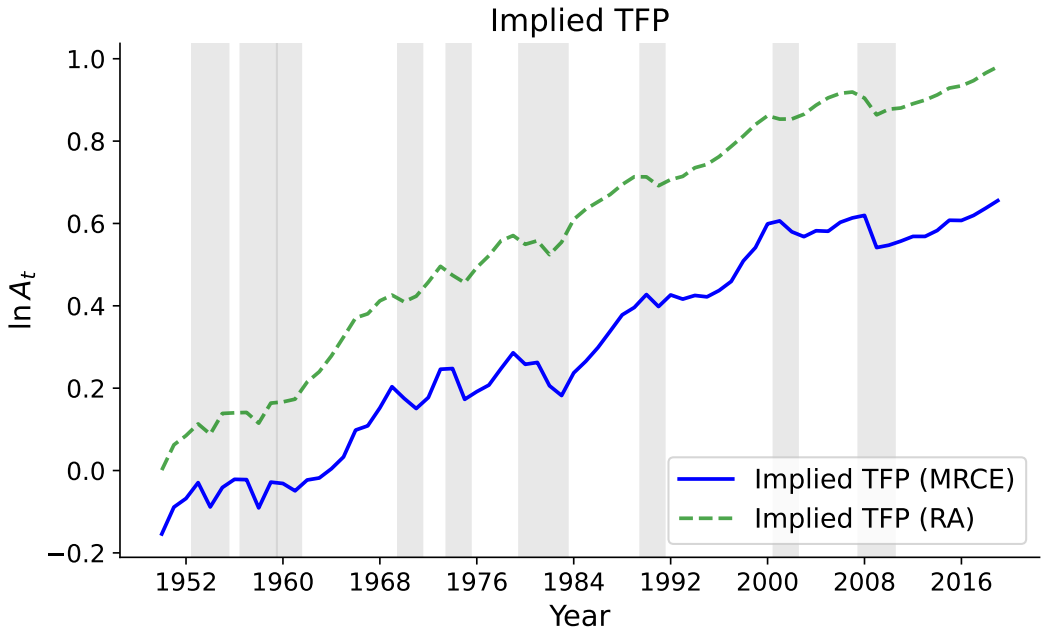
Table E.2 reports sensitivity to alternative calibrations. Approximate aggregation is extremely robust: I^{rep} accounts for 93–99 percent of total investment across all configurations. Near-unit-root persistence is robust to all parameters except the transfer share η.

Figure C.1: Return risk premium sensitivity



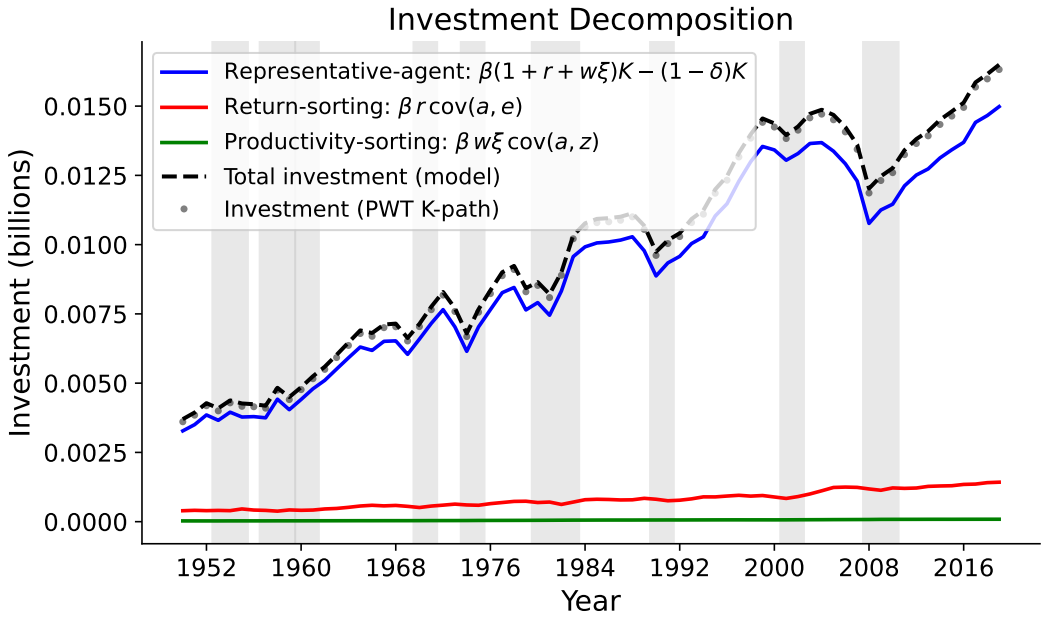
Notes: Left: static vs. dynamic premium components at steady state. Center: cumulative return differential over 30 years. Right: premium as a function of return dispersion σ_e.

Figure E.1: Implied TFP comparison



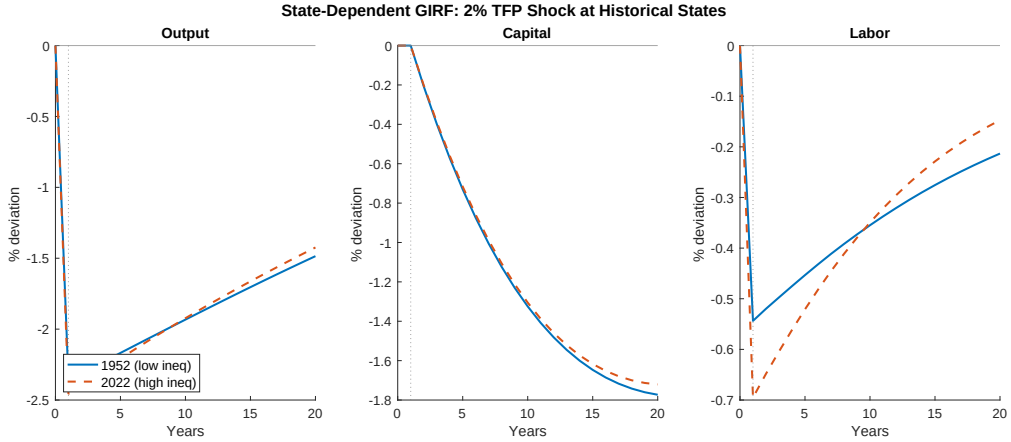
Notes: Implied TFP from the MRCE compared to the representative-agent analogue and the Fernald utilization-adjusted TFP series. Shaded areas indicate NBER recession dates.

Figure E.2: Decomposition of aggregate investment into three channels



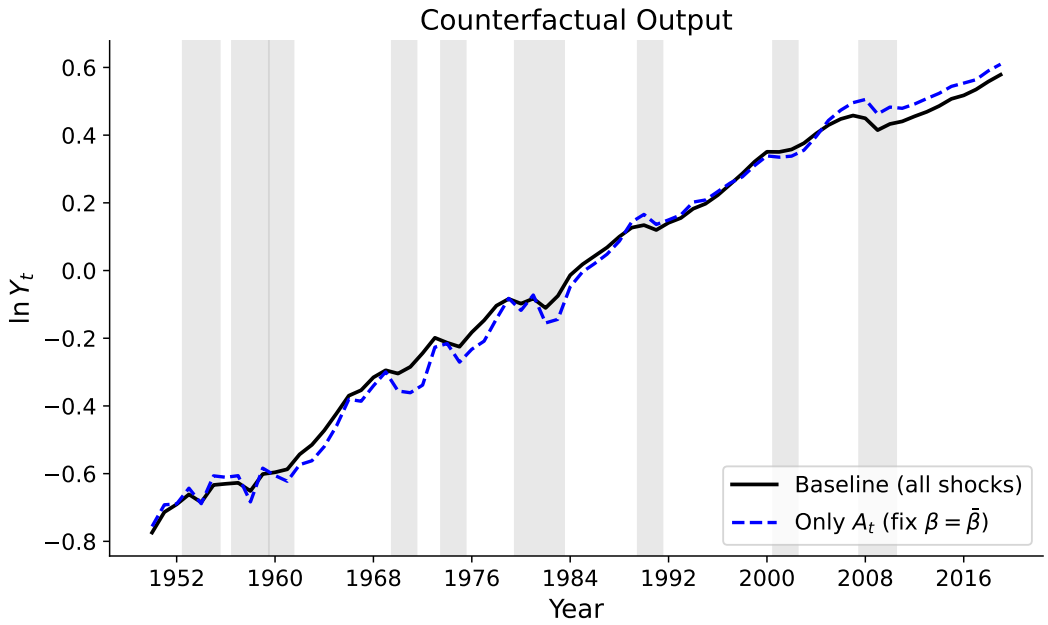
Notes: The three channels are the representative-agent channel (I^{rep}), the return-sorting channel (I^{ret}), and the productivity-sorting channel (I^{prod}). The representative-agent channel accounts for over 99.9% of mean investment. Gray dots show measured NIPA investment for reference; the gap reflects model abstractions (inventories, government and residential investment). Shaded areas indicate NBER recession dates.

Figure E.3: State-dependent GIRFs to a 2% negative TFP shock



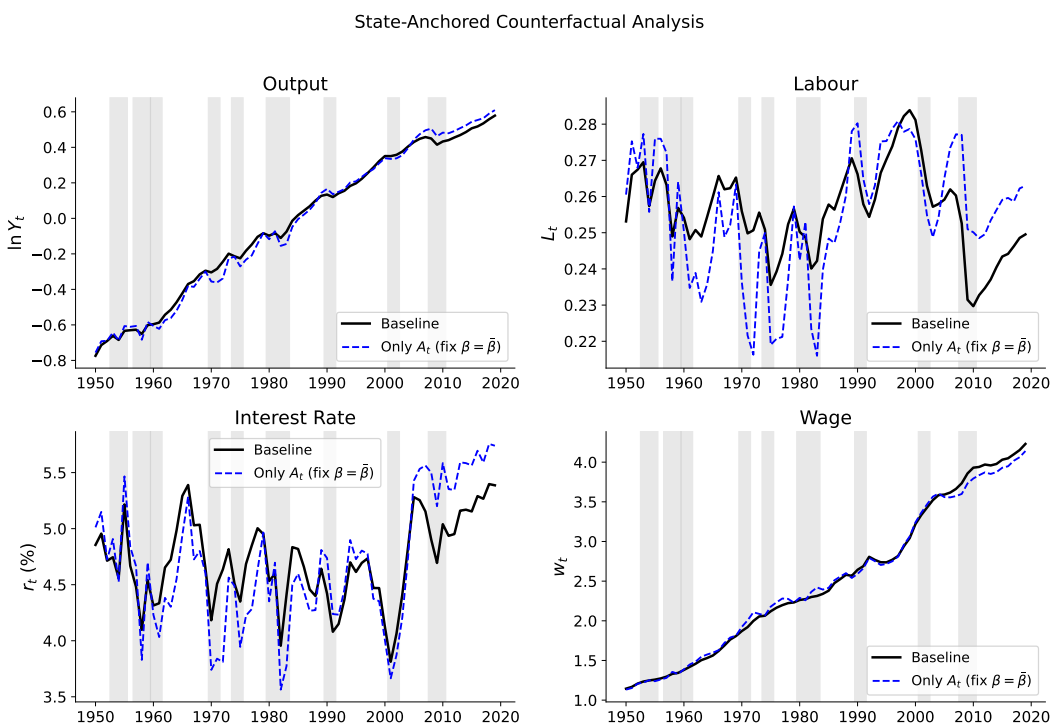
Notes: Averaged over 1,000 stochastic TFP paths. Solid: launched from the 1952 distributional state (low inequality). Dashed: launched from the 2019 distributional state (high inequality). The dotted vertical line marks the shock arrival.

Figure E.4: Counterfactual output paths



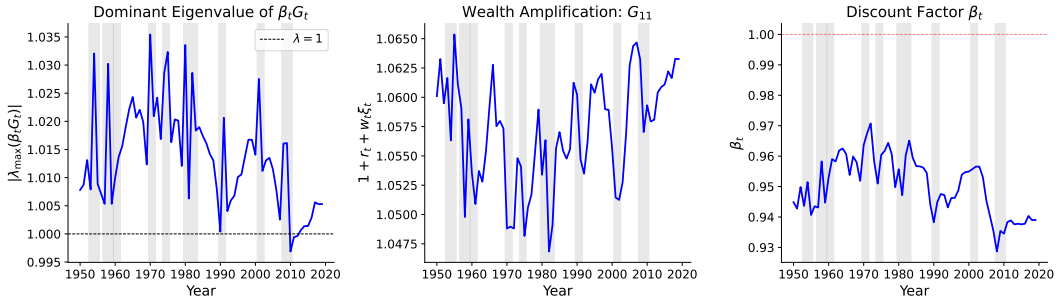
Notes: Baseline: all structural shocks at historical values. Fix β: β_t fixed at sample mean $\bar{\beta} = 0.954$, only $(A_t, Z_{L,t})$ vary. Fix Z_L: $Z_{L,t}$ fixed at sample mean $\bar{Z}_L = 1.08$, only (A_t, β_t) vary. Fix both: only A_t varies. Shaded areas indicate NBER recession dates.

Figure E.5: Counterfactual comparison across macroeconomic variables



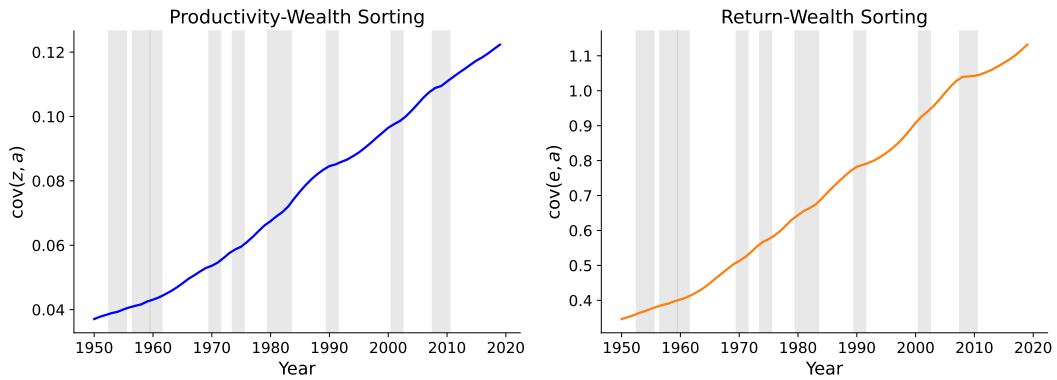
Notes: Panels show output (log), labor supply, interest rate, and wage. Each panel compares the baseline with the fix-β counterfactual. Shaded areas indicate NBER recession dates.

Figure E.6: Transition dynamics



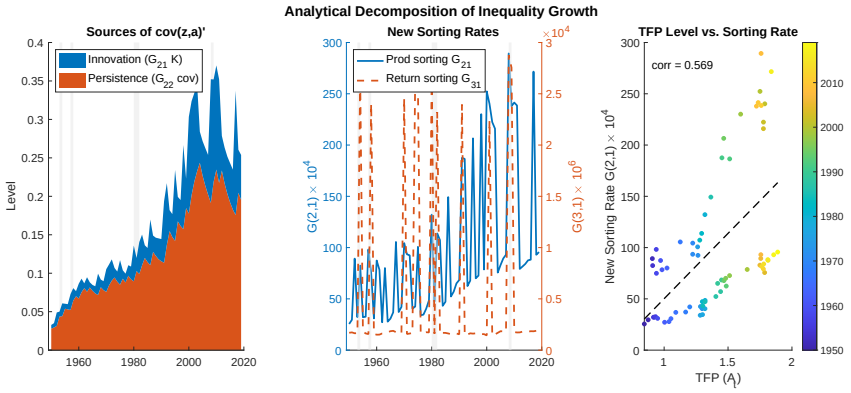
Notes: Top: dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$. Middle: wealth amplification factor. Bottom: time-varying discount factor. Shaded areas indicate NBER recession dates.

Figure E.7: Distributional state dynamics, 1950–2019



Notes: Top row: raw covariances. Bottom row: capital-normalized ratios. Shaded areas indicate NBER recession dates.

Figure E.8: Analytical decomposition of distributional sorting



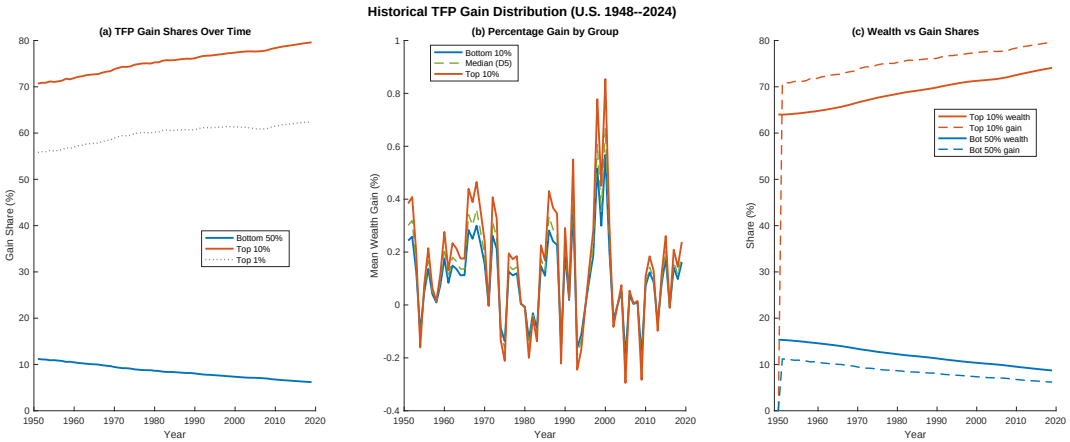
Notes: Left: innovation and persistence contributions. Center: sorting rate time paths. Right: TFP vs. sorting rate scatter.

Table E.1: Share of TFP-driven wealth gain by wealth decile (%), decade averages

Decade	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Top 5%	Top 1%
1950s	1.5	1.9	2.2	2.5	2.8	3.2	3.8	4.6	6.3	71.2	66.3	56.3
1960s	1.3	1.7	2.0	2.3	2.7	3.1	3.6	4.5	6.2	72.7	67.8	57.8
1970s	1.2	1.5	1.8	2.1	2.4	2.8	3.4	4.2	5.9	74.6	69.8	59.7
1980s	1.0	1.4	1.7	2.0	2.3	2.7	3.2	4.1	5.9	75.7	70.9	60.6
1990s	0.9	1.3	1.5	1.8	2.1	2.6	3.1	4.0	5.8	76.8	71.9	61.2
2000s	0.8	1.2	1.4	1.7	2.0	2.4	3.0	3.9	5.9	77.7	72.6	61.1
2010s	0.7	1.0	1.3	1.5	1.9	2.3	2.8	3.7	5.7	79.0	74.0	62.0
Full sample	1.1	1.4	1.7	2.0	2.3	2.7	3.3	4.1	5.9	75.4	70.5	59.9

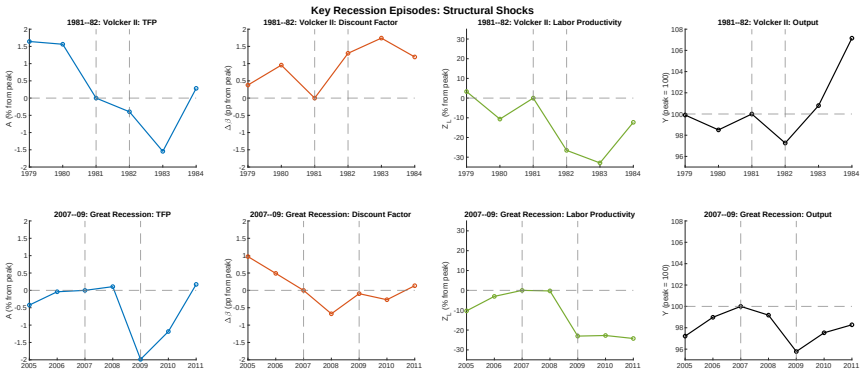
Notes: Each entry reports the average share (in %) of aggregate TFP-driven wealth gain captured by the indicated wealth decile. Period-by-period counterfactual at each year t.

Figure E.9: Historical distribution of TFP gains across wealth percentiles



Notes: Period-by-period counterfactual decomposition showing the share of TFP-driven wealth gains by wealth percentile, U.S. 1950–2019.

Figure E.10: Structural recession classification: shock paths around two contrasting episodes



Notes: 1981–82 (Volcker II, labor-driven) and 2007–09 (Great Recession, TFP/Labor). Each row shows one episode. Panels show TFP (A_t, log change from peak), discount factor (β_t, pp from peak), labor productivity mean ($Z_{L,t}$, log deviation from peak), and indexed output (Y_t, peak = 100). See Table 6 for the full classification.

Table E.2: Sensitivity of steady-state properties to alternative calibrations

Parameter	Value	K/Y	r (%)	$\lambda_{\max}$	I^{rep} (%)	Half-life (yr)
Baseline		3.38	2.77	0.9992	94.4	827
α	0.3	3.02	2.93	0.9991	94.0	749
	0.36	3.77	2.54	0.9992	94.8	905
	0.4	4.41	2.07	0.9993	95.7	1001
σ_e	0.001	3.24	3.17	0.9988	97.8	573
	0.005	3.29	3.02	0.9990	96.5	695
	0.02	3.54	2.32	0.9993	90.7	973
	0.05	3.86	1.54	0.9994	84.3	1157
	0.1	4.13	0.98	0.9995	79.9	1276
θ	1	3.38	2.77	0.9992	94.4	827
	1.5	3.38	2.77	0.9992	94.4	827
	3	3.38	2.77	0.9992	94.4	827
	5	3.38	2.77	0.9992	94.4	827
η	0.005	3.35	2.84	0.9997	95.0	2251
	0.01	3.37	2.79	0.9994	94.6	1110
	0.02	3.40	2.70	0.9987	93.8	541
	0.03	3.43	2.61	0.9980	93.1	353
	0.05	3.51	2.41	0.9966	91.4	204
β	0.94	2.62	5.62	0.9991	88.6	738
	0.95	2.95	4.19	0.9991	91.5	790
	0.97	3.98	1.29	0.9992	97.0	816
	0.98	5.03	-0.44	0.9992	98.1	903
ρ_z	0.9	3.37	2.79	0.9992	94.5	837
	0.93	3.37	2.78	0.9992	94.5	834
	0.97	3.40	2.71	0.9991	93.9	808
	0.99	3.65	2.04	0.9991	88.4	794

Notes: Each row varies one parameter from the baseline calibration.

References

Bach, L., L. E. Calvet, and P. Sodini (2020). Rich pickings? Risk, return, and skill in household wealth. *American Economic Review 110*(9), 2703–2747.

Bloom, N., M. Floetotto, N. Jaimovich, I. Saporta-Eksten, and S. J. Terry (2018). Really Uncertain Business Cycles. *Econometrica 86*(3), 1031–1065.

Brock, W. A. and L. J. Mirman (1972). Optimal economic growth and uncertainty: The discounted case. *Journal of Economic Theory 4*(3), 479–513.

Caballero, R. J. and M. L. Hammour (1994). The cleansing effect of recessions. *American Economic Review 84*(5), 1350–1368.

Chari, V. V., P. J. Kehoe, and E. R. McGrattan (2007). Business cycle accounting. *Econometrica 75*(3), 781–836.

Constantinides, G. M. and D. Duffie (1996). Asset pricing with heterogeneous consumers. *Journal of Political Economy 104*(2), 219–240.

Fagereng, A., L. Guiso, D. Malacrino, and L. Pistaferri (2020). Heterogeneity and persistence in returns to wealth. *Econometrica 88*(1), 115–170.

Kaplan, G., B. Moll, and G. L. Violante (2018). Monetary policy according to HANK. *American Economic Review 108*(3), 697–743.

Kesten, H. (1973). Random difference equations and renewal theory for products of random matrices. *Acta Mathematica 131*, 207–248.

Koop, G., M. H. Pesaran, and S. M. Potter (1996). Impulse Response Analysis in Nonlinear Multivariate Models. *Journal of Econometrics 74*(1), 119–147.

Krusell, P. and A. Smith, Jr. (1998, October). Income and Wealth Heterogeneity in the Macroeconomy. *Journal of Political Economy 106*(5), 867–896.

Piketty, T. (2014). *Capital in the Twenty-First Century*. Cambridge, MA: Harvard University Press.

Saez, E. and G. Zucman (2016). Wealth inequality in the United States since 1913: Evidence from capitalized income tax data. *Quarterly Journal of Economics 131*(2), 519–578.

Storesletten, K., C. I. Telmer, and A. Yaron (2004). Cyclical dynamics in idiosyncratic labor market risk. *Journal of Political Economy 112*(3), 695–717.